

Paper A-01: Earth Unified Environmental Transport: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

Founder, Vura Labs

Kampala, Uganda

ORCID: 0009-0001-0556-5516

May 2026

Abstract

This opening Earth-systems paper states the Part A modelling frame. It treats environmental transport as a CDFD/AFL hypothesis in which heat, water, nutrients, sediment, pollutants, and biological energy are represented as domain-specific fluxes Φ moving against constraints C through surfaces with responsiveness S and structural memory M_s . The release-local runtime outputs are internal diagnostics and tests that could break the mapping, not field evidence of climate, hydrological, ecological, or geological claims.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow-constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and

medicine [5]. Part IV [6], DOI <https://doi.org/10.5281/zenodo.20395901>, uses this Earth-systems opening paper to set the measurement boundary for the rest of Part A.

2 Earth-System Transport Frame

Earth systems are coupled transport networks. Atmospheric circulation moves heat and moisture; oceans redistribute heat and salinity; rivers move water and sediment; soils regulate nutrients and organic matter; ecosystems route biomass and information through trophic structure. CDFD/AFL does not replace the standard equations in these fields. It supplies a common audit layer for asking where flow rises, where capacity tightens, where responsiveness is lost, and where memory keeps a system from returning to its earlier state.

3 Candidate Variable Map

CDFD term	Earth-system reading	Example proxy
Φ	environmental drive or throughput	heat flux, runoff, nutrient load
C	active constraint or capacity burden	aridity, channel resistance, toxicity
S	responsiveness or buffering	biodiversity, soil structure, wetland storage
M_s	structural memory	degradation, hysteresis, legacy contamination
Ψ_s	operating ratio	balanced, constrained, or overloaded regime

This map is hypothesis-level until the proxies, scales, datasets, and failure conditions are specified for a given Earth-system question.

4 Part A Sequence

The Part A papers move from broad environmental transport into atmosphere, ocean, hydrology, soil, ecology, biodiversity, pollution, agriculture, sustainability, collapse, synthesis, plate tectonics, and volcanology. The sequence is ordered to start with transport structure, then add specific environmental constraints and failure modes.

5 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

6 Current Runtime Diagnostic

The release-local discovery script includes selected Earth-system adapter rows for climate and atmosphere, ocean transport, hydrological flow, biodiversity, and pollution. The accompanying network-cascade diagnostic checks finite-value behavior in an overload and memory-locking stress test. I use those rows as a runtime audit, not as a replacement for Earth-system data.

7 Falsification Targets

The Earth-systems framing weakens if a domain cannot define measurable proxies for Φ , C , S , and M_s , if the proposed operating ratio fails to predict overload or recovery behavior, or if a standard domain model explains the same data without any added value from the CDFD/AFL mapping.

8 Major Universal Upgrade: Mechanism, Scale, and Tests

8.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Earth Unified Environmental Transport, the proposed drive is radiative, hydrological, biogeochemical, and human forcing. The active limitation is transport resistance and finite environmental capacity. Responsiveness is carried by circulation, storage, ecological buffering, and rerouting, while retained state is represented by soil, ice, ocean, ecological, and contamination legacies. The outcome to explain is redistribution, overload, recovery, and regime change.

Table 1: Operational CDFD/AFL map for Paper A-01.

Term	Paper-specific observable meaning	Measurement question
Φ	radiative, hydrological, biogeochemical, and human forcing	What rate, load, gradient, or demand enters the focal system?
C	transport resistance and finite environmental capacity	Which measured bottleneck limits transfer, function, or recovery?
S	circulation, storage, ecological buffering, and rerouting	Which process changes effective capacity or reroutes the load?
M_s	soil, ice, ocean, ecological, and contamination legacies	Which retained state makes equal present loads produce different futures?
Y	redistribution, overload, recovery, and regime change	Which outcome is predicted out of sample, with uncertainty?

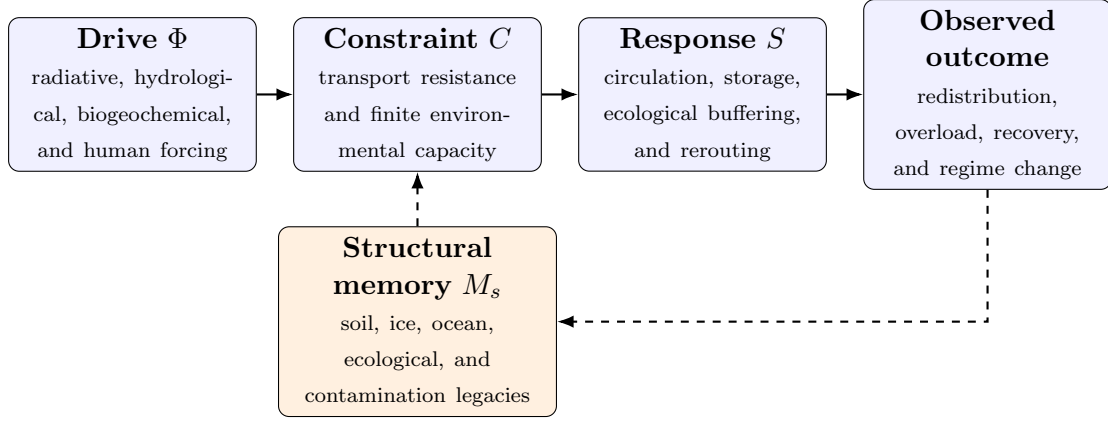


Figure 1: Paper A-01 observable CDFD/AFL architecture for Earth Unified Environmental Transport. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

8.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \quad (1)$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \quad (2)$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \quad (3)$$

$$\Psi_s(t) = \frac{\hat{\Phi}(t)}{\epsilon + \hat{C}(t)} \hat{S}(t) [1 + \omega \hat{M}_s(t)]. \quad (4)$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in radiative, hydrological, biogeochemical, and human forcing arrives faster than circulation, storage, ecological buffering, and rerouting can compensate. This raises or redistributes transport resistance and finite environmental capacity. If the load persists, the retained state encoded by soil, ice, ocean, ecological, and contamination legacies changes the next response, so a later event of equal magnitude can produce a different trajectory in redistribution, overload, recovery, and regime change. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

8.3 Cross-Scale Universality Without Mechanistic Erasure

For the Earth systems family, the universal comparison is between coupled transport, finite environmental capacity, buffering, and retained landscape or ecosystem state. Universality here cannot erase conservation laws, geometry, or scale; it must expose where

those domain equations create comparable overload and recovery structures. [2, 8, 1] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is hours to millennia; catchments to the coupled Earth system. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

8.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper A-01. Each rung can fail independently.

Test	Design	Discriminating result
Proxy audit	Measure coupled reanalysis, remote sensing, gauge records, and ecological time series and preregister how each record maps to Φ , C , S , M_s , and Y .	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a compound heat, rainfall, and pollutant pulse while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper’s stated falsifier is: the added variables do not improve prediction beyond ordinary coupled transport budgets.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ , separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

8.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from coupled reanalysis, remote sensing, gauge records, and ecological time series and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of redistribution, overload, recovery, and

regime change. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

9 Conclusion

Part A uses CDFD/AFL as a shared transport language for Earth systems. The framework is useful only when it sharpens measurable hypotheses and failure conditions. It remains separate from operational climate, hydrological, ecological, agricultural, or geological advice.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

10 Reference Layer

References

- [1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: <https://doi.org/10.1103/PhysRevLett.59.381>.
- [2] C. S. Holling. Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4:1–23, 1973. DOI: <https://doi.org/10.1146/annurev.es.04.110173.000245>.
- [3] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20250821>.
- [4] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20264779>.

- [5] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20344695>.
- [6] Steve Bico Mujjabi. Cdfd part iv: Universal adaptive flux limitation - constraint-driven flux dynamics, 2026. DOI: <https://doi.org/10.5281/zenodo.20395901>.
- [7] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdf execution engine, 2026. DOI: <https://doi.org/10.5281/zenodo.20343160>.
- [8] A. M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B*, 237(641):37–72, 1952. DOI: <https://doi.org/10.1098/rstb.1952.0012>.